

## 2 Kac's propagation of chaos

This section is mainly based on the Saint-Flour's lecture of Sznitman [Szn91], and on the recent surveys of Chaintron and Diez [CD22a, CD22b]. Please note that we refer in this document to the arXiv versions of these recent papers.

Let  $\mathcal{P}(E)$  be the set of probability measures on a Polish space  $E$  (separable completely metrizable topological space). The space of bounded and continuous functions on  $E$  is denoted by  $\mathcal{C}_b(E)$ . For  $\mu \in \mathcal{P}(E)$  and  $\varphi$  a test function,  $\langle \mu, \varphi \rangle = \int_E \varphi d\mu$ .

Let  $N \geq 2$  and consider a system  $\mathbf{X}^N = (X^{1,N}, \dots, X^{N,N}) \in E^N$  of  $N$  random interacting particles. We say **interacting particles** by opposition to independent variables.

In the interacting particle systems presented in Section 1, the particles are random processes defined on  $\mathbb{R}_+$ :  $X^{i,N} = (X_t^{i,N})_{t \geq 0}$ , which are either continuous or càdlàg. Their state spaces are therefore either  $E = \mathcal{C}(\mathbb{R}_+, \mathbb{R}^d)$  or  $E = \mathbb{D}(\mathbb{R}_+, \mathbb{R}^d)$ .

**Definition 2.1.** A particle system  $\mathbf{X}^N$  is said **exchangeable** (also called *symmetric*) when for any  $N \geq 2$ , the distribution of  $(X^{i,N})_{1 \leq i \leq N}$  is invariant by permutation of the indexes: for any  $\pi$  permutation of  $\{1, \dots, N\}$ ,  $(X^{1,N}, \dots, X^{N,N})$  and  $(X^{\pi(1),N}, \dots, X^{\pi(N),N})$  have the same distribution.

The exchangeability implies in particular that  $X^{1,N}$  has the same distribution as any particle  $X^{k,N}$ . The exchangeability assumption on the system  $\mathbf{X}^N$  is crucial for establishing propagation of chaos for an interacting particle system, that is, for reducing the study of the system in the large-population limit to the study of a single particle under a suitable distribution.

Provided that their initial configuration  $(X_0^{i,N})_{1 \leq i \leq N}$  is exchangeable, we observe that the interacting particle systems presented in Section 1 are exchangeable, since the interactions depend only on symmetric functionals of the particles, such as the empirical mean  $\frac{1}{N} \sum_{i=1}^N X^{i,N}$ , the empirical measure  $\frac{1}{N} \sum_{i=1}^N \delta_{X^{i,N}}$ , or the empirical cumulative distribution function  $\frac{1}{N} \sum_{i=1}^N \mathbb{1}_{X^{i,N} \leq x}$ .

**Definition 2.2.** Let  $\mu \in \mathcal{P}(E)$ . For each  $N \geq 2$ , let  $\mu^N$  denote the distribution of the system  $\mathbf{X}^N$ . We say that the system  $\mathbf{X}^N$  is  **$\mu$ -chaotic** if for any  $k \geq 1$  and any  $\varphi_1, \dots, \varphi_k$  in  $\mathcal{C}_b(E)$ ,

$$\begin{aligned} \lim_{N \rightarrow \infty} \langle \mu^N, \varphi_1 \otimes \dots \otimes \varphi_k \otimes 1 \otimes \dots \otimes 1 \rangle &= \lim_{N \rightarrow \infty} \mathbb{E}[\varphi_1(X^{1,N}) \dots \varphi_k(X^{k,N})] \\ &= \prod_{i=1}^k \langle \mu, \varphi_i \rangle. \end{aligned}$$

We note that if  $(Y_1, \dots, Y_k)$  are independent random variables with common distribution  $\mu$ , then the system  $\mathbf{X}^N$  is  $\mu$ -chaotic if, for any  $k \geq 1$  and any  $\varphi_1, \dots, \varphi_k$ ,

$$\lim_{N \rightarrow \infty} \mathbb{E}[\varphi_1(X^{1,N}) \dots \varphi_k(X^{k,N})] = \prod_{i=1}^k \mathbb{E}[\varphi_i(Y_i)] = \mathbb{E}[\varphi_1(Y_1) \dots \varphi_k(Y_k)].$$

Therefore, in the large-population limit ( $N \rightarrow \infty$ ), we observe that  $k$  interacting particles of the system behave as  $k$  independent particles with common distribution  $\mu$ .

## 2.1 Criterion for propagation of chaos

We now present a useful criterion for proving that a system is chaotic.

**Proposition 2.3.** *Let  $\mathbf{X}^N$  be an exchangeable system.*

*The system  $\mathbf{X}^N$  is  $\mu$ -chaotic if and only if its empirical distribution  $\bar{\mu}^N = \frac{1}{N} \sum_{i=1}^N \delta_{X^{i,N}}$  converges in law, as a  $\mathcal{P}(E)$ -valued random variable, to the deterministic probability measure  $\mu$  when  $N \rightarrow \infty$ .*

This result can be seen as a sort of Law of Large Numbers. If  $\mathbf{X}^N$  is  $\mu$ -chaotic, thus for any  $\varphi \in \mathcal{C}_b(E)$ ,

$$\begin{aligned} \langle \bar{\mu}^N, \varphi \rangle - \mathbb{E}[\varphi(X^{1,N})] &= \frac{1}{N} \sum_{i=1}^N \varphi(X^{i,N}) - \mathbb{E}[\varphi(X^{1,N})] \\ &\xrightarrow{N \rightarrow \infty} 0. \end{aligned}$$

Before proving Proposition 2.3, let us say a few words on the convergence in law of a sequence of random measures. The convergence in law is defined as the convergence against bounded continuous test functions. In the case of random measures, this representation is not intuitive.

However, we have the following useful result (proved in Section 2.2).

**Proposition 2.4** (Corollary 1 in [CD22a]). *Let  $(\mu^N)$  be a sequence of random probability measures. Consider  $\mu$  a deterministic probability measure, i.e.  $\text{Law}(\mu) = \delta_\mu$  with  $\mu \in \mathcal{P}(E)$ . There is equivalence between the following assertions*

- (i)  $(\mu^N)$  converge in law to  $\mu$ ,
- (ii)  $\lim_{N \rightarrow \infty} \mathbb{E}[|\langle \mu^N - \mu, \varphi \rangle|] = 0$  for any  $\varphi \in \mathcal{C}_b(E)$  uniformly continuous.

Note that, it is also equivalent to the convergence of  $\mathbb{E}[\langle \mu^N - \mu, \varphi \rangle^2]$  to 0 for any  $\varphi \in \mathcal{C}_b(E)$  uniformly continuous, since

$$\mathbb{E}[\langle \mu^N - \mu, \varphi \rangle^2] \leq 2\|\varphi\|_\infty \mathbb{E}[|\langle \mu^N - \mu, \varphi \rangle|] \leq 2\|\varphi\|_\infty \mathbb{E}[\langle \mu^N - \mu, \varphi \rangle^2]^{1/2}.$$

*Proof of Proposition 2.3.* This proposition is proved in [Szn91, Proposition 2.2] and [CD22a, Lemma 3.19].

1. Assume first that  $\mathbf{X}^N$  is  $\mu$ -chaotic. We prove the direct implication using Definition 2.2 with  $k = 1, 2$ .

Let  $\varphi \in \mathcal{C}_b(E)$ . We have

$$\begin{aligned} \mathbb{E}[\langle \bar{\mu}^N - \mu, \varphi \rangle^2] &= \mathbb{E}\left[\left(\frac{1}{N} \sum_{i=1}^N \varphi(X^{i,N}) - \langle \mu, \varphi \rangle\right)^2\right] \\ &= \frac{1}{N^2} \sum_{i,j=1}^N \mathbb{E}[\varphi(X^{i,N})\varphi(X^{j,N})] - \frac{2}{N} \langle \mu, \varphi \rangle \sum_{i=1}^N \mathbb{E}[\varphi(X^{i,N})] + \langle \mu, \varphi \rangle^2. \end{aligned}$$

Using the exchangeability of the system, we deduce

$$\begin{aligned}\mathbb{E}\left[\langle \bar{\mu}^N - \mu, \varphi \rangle^2\right] &= \frac{1}{N} \mathbb{E}[\varphi(X^{1,N})^2] \\ &\quad + \frac{N-1}{N} \mathbb{E}[\varphi(X^{1,N})\varphi(X^{2,N})] - 2\langle \mu, \varphi \rangle \mathbb{E}[\varphi(X^{1,N})] + \langle \mu, \varphi \rangle^2,\end{aligned}$$

which converges to 0 when  $N \rightarrow \infty$  by Definition 2.2. We deduce the convergence in law of the random measure  $\bar{\mu}^N$  to the constant measure  $\mu$  (by Proposition 2.4).

2. Conversely, assume that the random probability measures  $\bar{\mu}^N$  converges in law to the deterministic probability measure  $\mu$ .

Let  $k \geq 1$  and  $\varphi_1, \dots, \varphi_k$  in  $\mathcal{C}_b(E)$ . We have

$$\begin{aligned}\left| \langle \bar{\mu}^N, \varphi_1 \otimes \dots \otimes \varphi_k \otimes 1 \otimes \dots \otimes 1 \rangle - \prod_{i=1}^k \langle \mu, \varphi_i \rangle \right| &= \left| \mathbb{E}[\varphi_1(X^{1,N}) \dots \varphi_k(X^{k,N})] - \prod_{i=1}^k \langle \mu, \varphi_i \rangle \right| \\ &\leq \left| \mathbb{E}[\varphi_1(X^{1,N}) \dots \varphi_k(X^{k,N})] - \mathbb{E}\left[\prod_{i=1}^k \langle \bar{\mu}^N, \varphi_i \rangle\right] \right| + \left| \mathbb{E}\left[\prod_{i=1}^k \langle \bar{\mu}^N, \varphi_i \rangle\right] - \prod_{i=1}^k \langle \mu, \varphi_i \rangle \right|. \quad (2.1)\end{aligned}$$

By Proposition 2.4, since  $\bar{\mu}^N$  converges in law to  $\mu$ , and  $\varphi_i \in \mathcal{C}_b(E)$ , we have the convergence of  $\langle \bar{\mu}^N, \varphi_i \rangle$  to  $\langle \mu, \varphi_i \rangle$  in  $L^1$  for any  $i \in \{1, \dots, k\}$ . As the functions  $\varphi_i$  are bounded, we deduce by induction on  $k$  the convergence of the second term in (2.1) to 0 when  $N \rightarrow \infty$ .

We now study the first term of (2.1).

Using the exchangeability of the system, we note that

$$\mathbb{E}[\varphi_1(X^{1,N}) \dots \varphi_k(X^{k,N})] = \frac{1}{A(k, N)} \sum_{\substack{i_1, \dots, i_k \\ \text{pairwise distinct}}} \mathbb{E}[\varphi_1(X^{i_1,N}) \dots \varphi_k(X^{i_k,N})],$$

where  $A(k, N) = \frac{N!}{(N-k)!}$  is the number of pairwise distinct tuples  $(i_1, \dots, i_k)$  of integers between 1 and  $N$ . In addition, by definition of  $\bar{\mu}^N$ ,

$$\begin{aligned}\prod_{i=1}^k \langle \bar{\mu}^N, \varphi_i \rangle &= \frac{1}{N^k} \prod_{i=1}^k \left( \sum_{j=1}^N \varphi_i(X^{j,N}) \right) = \frac{1}{N^k} \sum_{i_1, \dots, i_k} \varphi_1(X^{i_1,N}) \dots \varphi_k(X^{i_k,N}) \\ &= \frac{1}{N^k} \sum_{\substack{i_1, \dots, i_k \\ \text{pairwise distinct}}} \varphi_1(X^{i_1,N}) \dots \varphi_k(X^{i_k,N}) + R(k, N),\end{aligned}$$

Denoting by  $C = \sup_{1 \leq i \leq k} \|\varphi_i\|_\infty$ , we note that the remainder term  $R(k, N)$  satisfies

$$|R(k, N)| \leq \frac{C^k}{N^k} (N^k - A(k, N)).$$

Consequently,

$$\begin{aligned}
& \left| \mathbb{E}[\varphi_1(X^{1,N}) \dots \varphi_k(X^{k,N})] - \mathbb{E}\left[\prod_{i=1}^k \langle \bar{\mu}^N, \varphi_i \rangle\right] \right| \\
& \leq \left( \frac{1}{A(k,N)} - \frac{1}{N^k} \right) A(k,N) C^k + \mathbb{E}[|R(k,N)|] \\
& \leq 2C^k \left( 1 - \frac{A(k,N)}{N^k} \right) \leq 2C^k \left( 1 - \left( 1 - \frac{k-1}{N} \right)^k \right) \leq 2C^k \frac{k(k-1)}{N},
\end{aligned}$$

which goes to 0 when  $N \rightarrow \infty$ . Then  $\mathbf{X}^N$  is  $\mu$ -chaotic. □

## 2.2 Auxiliary proofs

For completeness of this section, we now prove Proposition 2.4. Recall that weak convergence of measures in  $\mathcal{P}(\mathcal{P}(E))$  is defined as the convergence against test functions in  $\mathcal{C}_b(\mathcal{P}(E))$ . An example of such functionals is given by the linear functions:  $\mu \mapsto \langle \mu, \varphi \rangle$  for a given  $\varphi \in \mathcal{C}_b(E)$ . However, the weak convergence of measures is not simple to represent.

Note that for a random variable  $X$  on  $E$ , we usually denote by  $\mu \in \mathcal{P}(E)$  its distribution:  $\mu = \text{Law}(X)$ . For a  $\mathcal{P}(E)$ -valued random variable  $\mu$ , we denote here by  $\hat{\mu} \in \mathcal{P}(\mathcal{P}(E))$  its distribution:  $\hat{\mu} = \text{Law}(\mu)$ .

We introduce various distances on  $\mathcal{P}(E)$  and  $\mathcal{P}(\mathcal{P}(E))$ . First, the distance  $\mathcal{D}$  on  $\mathcal{P}(E)$ , defined by

$$\mathcal{D}(\mu, \nu) = \sum_{k=1}^{\infty} \frac{1}{2^k} |\langle \mu - \nu, \varphi_k \rangle|,$$

where  $(\varphi_k)_{k \in \mathbb{N}}$  are continuous bounded functions on  $E$  with  $\|\varphi_k\|_{\infty} \leq 1$ , and satisfying the *separating property* (see [EK86, Section III-4]):  $\forall \mu, \nu \in \mathcal{P}(E)$ ,

$$\left( \forall k \in \mathbb{N} \quad \langle \mu, \varphi_k \rangle = \langle \nu, \varphi_k \rangle \right) \implies \left( \mu = \nu \right).$$

Recall that  $E$  is a polish space, therefore such a dense and countable family  $(\varphi_k)_{k \in \mathbb{N}}$  exists. The distance  $\mathcal{D}$  metrizes the topology of weak convergence on  $\mathcal{P}(E)$ .

Set  $\mathcal{E} := \mathcal{P}(E)$ . For  $\hat{\mu} \in \mathcal{P}(\mathcal{E})$  and  $\hat{\nu} \in \mathcal{P}(\mathcal{E})$ , let  $\mathcal{W}_{\mathcal{D}}$  the Wasserstein distance on  $\mathcal{P}(\mathcal{E})$  defined by

$$\mathcal{W}_{\mathcal{D}}(\hat{\mu}, \hat{\nu}) := \inf_{\mu \sim \hat{\mu}, \nu \sim \hat{\nu}} \mathbb{E}[\mathcal{D}(\mu, \nu)] = \inf_{\hat{\pi} \in \Pi(\hat{\mu}, \hat{\nu})} \int \mathcal{D}(\mu, \nu) \hat{\pi}(d\mu, d\nu)$$

with  $\Pi(\hat{\mu}, \hat{\nu})$  is the set of probability measures  $\hat{\pi}$  on  $\mathcal{E} \times \mathcal{E}$  with marginals  $\hat{\mu}$  and  $\hat{\nu}$ .

We first prove the following result.

**Proposition 2.5** (Proposition 6 in [CD22a]). *Let  $(\mu^N)_{N \in \mathbb{N}}$  be a sequence of random probability measures and  $\mu$  be a random probability measure.*

(i) if  $\lim_{N \rightarrow \infty} \mathcal{W}_{\mathcal{D}}(\text{Law}(\mu^N), \text{Law}(\mu)) = 0$ , then the sequence  $(\mu^N)_{N \geq 0}$  of  $\mathcal{P}(E)$ -valued random variables converges in law towards  $\mu$ .

(ii) if  $\forall \varphi \in \mathcal{C}_b(E)$  uniformly continuous,  $\lim_{N \rightarrow \infty} \mathbb{E}[|\langle \mu^N - \mu, \varphi \rangle|] = 0$ , then

$$\lim_{N \rightarrow \infty} \mathcal{W}_{\mathcal{D}}(\text{Law}(\mu^N), \text{Law}(\mu)) = 0.$$

(iii) If for any  $\varphi \in \mathcal{C}_b(E)$  uniformly continuous, we have

$$\lim_{N \rightarrow \infty} \mathbb{E}[|\langle \mu^N - \mu, \varphi \rangle|] = 0,$$

then  $(\mu^N)_{N \in \mathbb{N}}$  converges in law to  $\mu$ .

*Proof.* (i) Recall that the convergence in law is characterized by the convergence of integrals against test functions in the set of bounded uniformly continuous functions (Portmanteau theorem, see [Bil99, Theorem 2.1]). Let  $\Phi \in \mathcal{C}_b(\mathcal{E})$  be a function uniformly continuous for the metric  $\mathcal{D}$ : for any  $\varepsilon > 0$ , there exists  $\delta := \delta(\varepsilon) > 0$  such that for any  $\mu, \nu \in \mathcal{E}$ ,

$$\mathcal{D}(\mu, \nu) \leq \delta \quad \Rightarrow \quad |\Phi(\mu) - \Phi(\nu)| \leq \varepsilon.$$

Recall the well-known relation for a random variable  $X$ :  $\langle \text{Law}(X), \Phi \rangle = \int \Phi d\mathbb{P}_X = \mathbb{E}[\Phi(X)]$ . Consequently, we have

$$\begin{aligned} |\langle \text{Law}(\mu^N) - \text{Law}(\mu), \Phi \rangle| &\leq \mathbb{E}[|\Phi(\mu^N) - \Phi(\mu)|] \\ &\leq \varepsilon + 2\|\Phi\|_{\infty} \mathbb{P}(|\Phi(\mu^N) - \Phi(\mu)| \geq \varepsilon) \\ &\leq \varepsilon + 2\|\Phi\|_{\infty} \mathbb{P}(\mathcal{D}(\mu^N, \mu) > \delta) \\ &\leq \varepsilon + \frac{2\|\Phi\|_{\infty}}{\delta} \mathbb{E}[\mathcal{D}(\mu^N, \mu)], \end{aligned} \tag{2.2}$$

where the uniform continuity of  $\Phi$  is used in the third line, and Markov's inequality in the last line. Taking the infimum on every pair of random measures  $(\mu^N, \mu)$  with distribution in  $\Pi(\text{Law}(\mu^N), \text{Law}(\mu))$ , we deduce

$$|\langle \text{Law}(\mu^N) - \text{Law}(\mu), \Phi \rangle| \leq \varepsilon + \frac{2\|\Phi\|_{\infty}}{\delta} \mathcal{W}_{\mathcal{D}}(\text{Law}(\mu^N), \text{Law}(\mu)).$$

The result of (i) follows by taking first the limit when  $N \rightarrow \infty$ , and then when  $\varepsilon \rightarrow 0$ . We thus recover the fact that the topology induced by Wasserstein distance is stronger than the one induced by the weak convergence.

(ii) By definition of  $\mathcal{W}_{\mathcal{D}}$  and the monotone convergence theorem, we have

$$\begin{aligned} \mathcal{W}_{\mathcal{D}}(\text{Law}(\mu^N), \text{Law}(\mu)) &\leq \mathbb{E}[\mathcal{D}(\mu^N, \mu)] \\ &\leq \sum_{k=1}^{\infty} \frac{1}{2^k} \mathbb{E}[|\langle \mu^N - \mu, \varphi_k \rangle|]. \end{aligned}$$

Using the dominated convergence theorem, the conclusion of (ii) follows.

(iii) if  $\forall \varphi \in \mathcal{C}_b(E)$  uniformly continuous,  $\lim_{N \rightarrow \infty} \mathbb{E}[|\langle \mu^N - \mu, \varphi \rangle|] = 0$ , (ii) implies  $\lim_{N \rightarrow \infty} \mathcal{W}_{\mathcal{D}}(\text{Law}(\mu^N), \text{Law}(\mu)) = 0$ , and thus, by (i),  $(\mu^N)_{N \geq 0}$  converges in law to  $\mu$ . The proposition is proved.  $\square$

We are now able to prove Proposition 2.4.

*Proof of Proposition 2.4.* The implication (ii)  $\Rightarrow$  (i) is given by Proposition 2.5. To prove the direct implication (i)  $\Rightarrow$  (ii), we observe that for a fixed  $\varphi \in \mathcal{C}_b(E)$ , when  $\mu$  is deterministic the function  $\Phi : \nu \mapsto |\langle \nu - \mu, \varphi \rangle|$  is a deterministic function on  $\mathcal{P}(E)$ . In addition,  $\Phi$  is bounded by  $2\|\varphi\|_{\infty}$  and continuous for the weak topology on  $\mathcal{P}(E)$ :

$$|\Phi(\nu_1) - \Phi(\nu_2)| = ||\langle \nu_1 - \mu, \varphi \rangle| - |\langle \nu_2 - \mu, \varphi \rangle|| \leq |\langle \nu_1 - \nu_2, \varphi \rangle|,$$

by the reverse triangle inequality. Consequently, it is a test function for the convergence in law, and since  $(\mu^N)$  converges in law to  $\mu$ , the result follows.  $\square$

In Section 1, several interacting particle systems were introduced, where  $\mathbf{X}^N = (X_t^{1,N}, \dots, X_t^{N,N})_{t \geq 0}$  are càdlàg processes with values in  $\mathbb{R}^d$ ,  $d \geq 1$ . In the next section, we recall some properties of weak convergence and tightness for such processes. In Sections 4 and 5, we analyze two examples in details and, whenever possible, provide an estimate of the rate of convergence to the asymptotic distribution. The long-time behavior of the system and of its asymptotic distribution are also studied.